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RESEARCH

FOCUS: Frequency-Optimized Conditioning of diffUSion models for mitigating catastrophic forgetting during test-time adaptation

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Abstract

Test-time adaptation enables models to adapt to evolving domains. However, balancing the tradeoff between preserving knowledge and adapting to domain shifts remains challenging for model adaptation methods, since adapting to domain shifts can induce forgetting of task-relevant knowledge. To address this problem, we propose FOCUS, a novel frequency-based conditioning approach within a diffusion-driven input-adaptation framework. Utilising learned, spatially adaptive frequency priors, our approach conditions the reverse steps during diffusion-driven denoising to preserve task-relevant semantic information for dense prediction. FOCUS leverages a trained, lightweight, Y-shaped Frequency Prediction Network (Y-FPN) that disentangles high and low frequency information from noisy images. This minimizes the computational costs involved in implementing our approach in a diffusion-driven framework. We train Y-FPN with FrequencyMix, a novel data augmentation method that perturbs the images across diverse frequency bands, which improves the robustness of our approach to diverse corruptions. We demonstrate the effectiveness of FOCUS for semantic segmentation and monocular depth estimation across 15 corruption types and three datasets, achieving state-of-the-art averaged performance. In addition to improving standalone performance, FOCUS complements existing model adaptation methods since we can derive pseudo labels from FOCUS denoised images for additional supervision. Even under limited, intermittent supervision from the derived pseudo labels, we show that FOCUS mitigates catastrophic forgetting for recent model adaptation methods. Source code is available at https://github.com/gabriel-tjio/FOCUS.

Keywords Test-time adaptation · Diffusion · Dense prediction · Frequency-aware modelling

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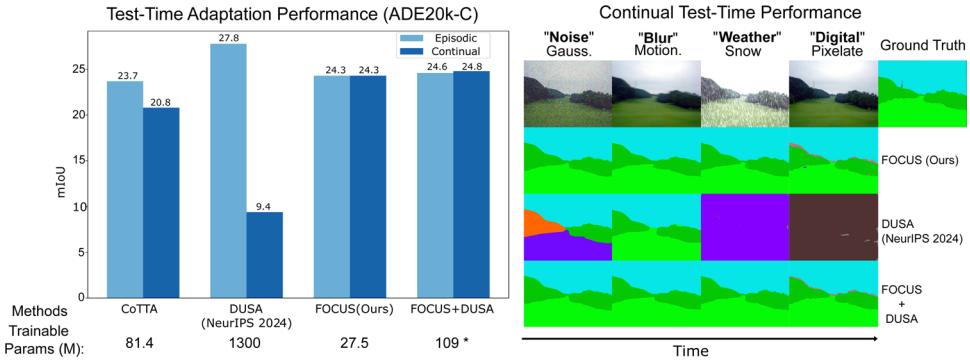

Fig. 1 Performance on ADE20k-C. Left: Averaged mIoU across 15 corruptions for episodic (model resets after each corruption) and continual (adapts model across all corruptions without reset) settings. As a denoising approach, FOCUS is not affected by catastrophic forgetting and recovers degraded performance for DUSA 'FOCUS+DUSA'. Right: Continual test-time qualitative results on representative corruptions from ADE20k-C. *: DUSA, a diffusion-based method, updates ControlNet, UNet and the task model (Segformer-b5), which raises the number of trainable parameters. For 'FOCUS+DUSA', we only update the task model during adaptation. For 'FOCUS', we only optimize the parameters for FOCUS before adaptation and do not update the task model

1 Introduction

Deep learning models often suffer from significantly reduced performance when deployed in the wild, where test time data distribution diverges from the source domain (training data) distribution [1, 2]. To address distribution shifts, model adaptation [3–6], which improves test-time performance through self-supervised fine-tuning, has emerged as a compelling solution. However, catastrophic forgetting [5], where adapting to the test data degrades existing task-relevant knowledge, presents a considerable challenge for model adaptation.

Catastrophic forgetting occurs when the test data distribution varies over time, and fine tuning the task model with the test data may result in the loss of the original knowledge from the pretrained source model. While this can be mitigated by preserving the existing knowledge by limiting updates of the task model [5] during test-time adaptation, performance drops noticeably under continual settings (Fig. 1). In Fig. 1, catastrophic forgetting erodes performance over time as the model overfits to the test data in each episode, introducing a gap between episodic and continual adaptation performance.

Model adaptation methods generally assume that pseudo label quality is acceptable for model finetuning. However, this assumption may not hold under extreme conditions. Input adaptation methods [7–9], which transform the target domain data to match the source domain data, present a possible solution to improve pseudo label quality. Prior work [7] has demonstrated the effectiveness of mapping multiple target domains back to the source data for image classification. The effectiveness of conditioning diffusion models [10–15] in generating user-defined outputs has motivated their use for image denoising [7, 11]. However, these methods use predefined, image-level priors for conditioning, which may be insufficient for dense prediction.

1.1 Our approach

In this work, we introduce learned, spatially adaptive pixel-wise kernels to enhance the effectiveness of diffusion-driven denoising for dense prediction. We hypothesize that the semantic information necessary for good performance in dense prediction tasks comprise high frequency (*e.g.* edges) and low frequency (*e.g.* colour) signals, leading us to propose a simple yet effective context-aware frequency prediction network to preserve semantic information. Since it is non-trivial to manually define robust, generalizable frequency priors, we propose to train a Y-shaped Frequency Prediction Network(Y-FPN) that extracts frequency priors from noisy images. We then use these priors to condition diffusion-driven denoising for dense prediction (segmentation and depth estimation). To the best of our knowledge, this is the first work to condition diffusion denoising via learned pixel-level frequency priors for dense prediction tasks. We list our contributions below:

- Our novel workflow, Frequency-Optimized Conditioning in DiffUSion (FOCUS) introduces a novel approach that mitigates catastrophic forgetting by using spatially adaptive frequency priors to condition a pretrained diffusion model for diffusion-driven denoising. We extract the learned frequency priors with a lightweight Y-shaped Frequency Prediction Network (Y-FPN). To train Y-FPN, we introduce FrequencyMix to diversify training data.
- *Performance* We evaluate FOCUS on severe, diverse corruptions (n=15) from the ImageNet-C dataset and show that FOCUS outperforms best existing work on

averaged performance for segmentation- ADE20k-C: (DUSA [6]:+14.9% mIoU, DDA[7]:1.7 % mIoU); Cityscapes-C:(CoTTA [5]: +3.7% mIoU) and depth estimation -NYU2k-C: (DDA [7]:−0.32 relative error).

- *Complementarity* As a task agnostic, input adaptation method, FOCUS can be seamlessly combined with model adaptation techniques. FOCUS yields synergistic gains for both segmentation and depth estimation, validating its utility as a flexible, modular addition to TTA pipelines. Importantly, FOCUS benefits model adaptation even with intermittent supervision for semantic segmentation (*e.g.*(ADE20k-C) 'FOCUS+TENT (k=25)':+0.5% mIoU, where the pseudo labels from the denoised images are applied once every 25 iterations) and depth estimation: 'FOCUS+CoTTA (k=50)':−0.06 relative error), highlighting the effectiveness of FOCUS.

2 Related work

2.1 Test-time domain adaptation

Test-time adaptation can be categorized into *model adaptation* and *input adaptation*. Model adaptation methods finetune the model during inference Since the ground truth labels are unavailable at test time, model adaptation methods rely on self supervision approaches such as entropy minimization [4], pseudo labeling [5, 16], attention-based methods [17, 18] and feature distribution alignment [19–21]. Since the pseudo labels are noisy, much effort has been devoted to improving pseudo label reliability using methods such as consistency regularization [5], which encourages similar model output from augmented instances of the same input image or by pseudo label filtering [22]. Although numerous prior studies [4–6, 23–26] study TTA for dense prediction tasks, the majority of existing work finetune the models on the test data.

Input adaptation methods transform the styles of the target domain data to match the styles of the original training data (source domain images) while preserving semantic information of the target domain images. Earlier work [27–29] rely on Generative Adversarial Networks (GANs), which perform a one-to-one transformation from the target domain images to the source domain images. Unlike GAN-based approaches [27–29], diffusion models offer a greater degree of fine-grained control over the generated outputs [7–9, 30] and map the target domains to the source domain data (many-to-one transformation). Kim *et al.*[28] observed that arbitrary style transfer-based approaches introduce image artifacts in the generated images which can hinder adaptation performance. While our work updates the input (input adaptation) via image denoising, we also show that our work can be integrated with existing model adaptation methods to improve performance.

2.2 Diffusion models

Diffusion modelling has recently gained widespread prominence and demonstrates strong generative capacity [31–37]. The underlying concept for diffusion (*i.e.*denoising diffusion probabilistic model (DDPM) [31]) can be described as follows - the 'forward process' iteratively adds noise to the data, and the 'reverse process' leverages a trained network to progressively recover the original data through a series of steps.

Diffusion models show a high degree of user controllability via conditioning on visual cues [11], natural language [12, 13], and class labels [14, 15]. Recent work [7, 38, 39] demonstrated the effectiveness of diffusion models in projecting non-Gaussian corrupted data to the source distribution. In particular, DDA (Gao *et al.*[7]) show the effectiveness of a predefined image-level low-pass filter for robust test-time input adaptation of image classification tasks. Gao *et al.*[7] used predefined image-level filters to preserve the image-level semantic information of the corrupted images during denoising. We hypothesize that dense prediction tasks require pixel-level priors to preserve the semantic information during diffusion-driven denoising. Additionally, the learned priors must be robust against broad range of image corruptions, leading us to investigate frequency-based methods for generating training data.

2.3 Frequency-aware processing

There is substantial interest in applying frequency modelling to guide the development of robust, generalizable vision systems [7, 11, 40–42]. It was previously observed that classification models are biased towards colour and texture information [43], and that vision models make predictions based on a combination of low and high-frequency information [44, 45]. Additionally, prior studies [40, 46] show that the frequency spectrum presents a representational space to separate and extract the underlying patterns and semantic information from an image.

This is useful for data augmentation, especially since the goal is to increase data diversity without changing the semantic meaning of the images. Earlier work [40] preferentially perturb the high-frequency amplitude spectrum compared to the low-frequency amplitude spectrum of the training data to improve adaptation performance to real-world data (*i.e.* Synthetic training data → Real-world semantic segmentation datasets). However, since our goal to improve generalizability to a broad spectrum of noise and image corruptions, we train our model using a composite augmented

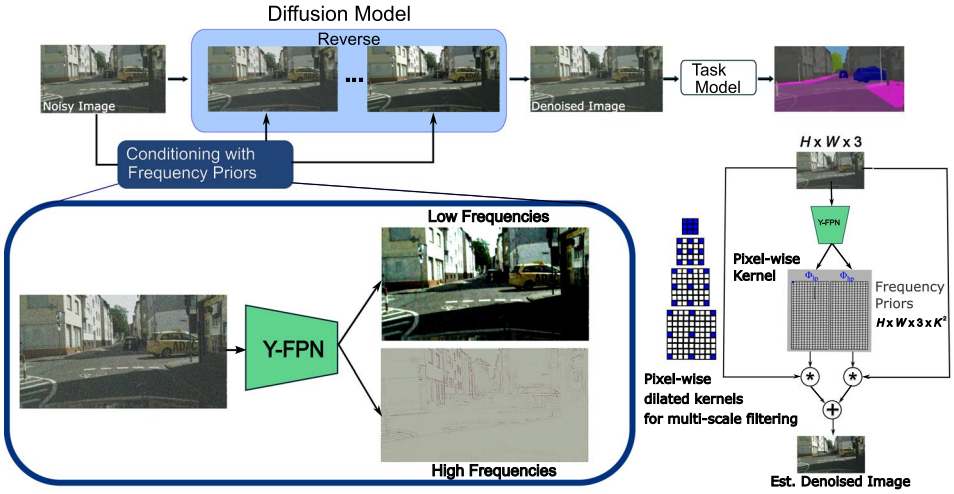

Fig. 2 FOCUS Workflow. We condition the reverse denoising steps with our spatially adaptive frequency priors. The Y-shaped Frequency Prediction Network (Y-FPN) outputs the frequency priors (Φ_{lp} and Φ_{hp}) of size $K \times K \times 3$ for each pixel of the input noisy RGB image with dimensions $H \times W \times 3$ pixels

image generated from our novel approach, FrequencyMix, which combines augmentations sampled from frequency space. Here, our objective is to generate sufficiently diverse noisy images by sampling in frequency space so that our trained model will be robust to noise at test time.

In addition to training robust models, frequency-aware processing can also be used to condition generative priors. Notably, Choi *et al*.[11] and Gao *et al*.[7] demonstrated the effectiveness of conditioning the diffusion model with predefined image-level low-pass filters for user-defined image generation. We extend their work and introduce learned, spatially adaptive low and high-frequency priors.

3 Methods

In this section, we introduce FOCUS (Fig. 2), a frequency-conditioned diffusion-driven input adaptation method. We first introduce the theoretical background in Sect. 3.1 for conditioning diffusion-driven denoising. In Sect. 3.2, we introduce our lightweight (27.5M parameters) Y-shaped Frequency Prediction Network (Y-FPN) which generates spatially adaptive frequency priors. We also show the loss functions used to train Y-FPN (Sect. 3.3) and the test-time loss functions (Sect. 3.4) used for additional supervision from the FOCUS denoised images.

3.1 Leveraging frequency priors to condition denoising via diffusion

Given an input image $\boldsymbol{x}_0$ randomly sampled from a target distribution $q(\boldsymbol{x}_0)$, we leverage a ImageNet pretrained diffusion model [14] to map $\boldsymbol{x}_0$ to the source domain.

Diffusion comprises a forward process, which iteratively adds Gaussian noise to $\boldsymbol{x}_0$ over N steps (N=1000), and a reverse process, which estimates the input image $\boldsymbol{x}_t^g$ for each step t. g indicates guidance (*e.g.*via frequency priors).

For step $t-1$, we sample the estimate $\hat{\boldsymbol{x}}_{t-1}^g$ from the conditional distribution, *i.e.*, $\hat{\boldsymbol{x}}_{t-1}^g \sim p_\theta(\boldsymbol{x}_{t-1}^g | \boldsymbol{x}_t^g)$, where p_θ is the parameterized learned process that approximates the reverse of the forward noising steps.

To preserve essential semantic information during diffusion-driven denoising for image classification, Gao *et al*.[7] introduce an image-level, low-frequency preserving constraint to the reverse process. For the time step $t-1$, Gao *et al*. estimate the denoised image $\hat{\boldsymbol{x}}_0^g$ by

$$\hat{\boldsymbol{x}}_0^g = \sqrt{\frac{1}{\overline{\alpha}_t}}\boldsymbol{x}_t^g - \epsilon_\theta(\boldsymbol{x}_t^g, t)\sqrt{\frac{1}{\overline{\alpha}_t} - 1}. \tag{1}$$

$\alpha_t := 1 - \beta_t$, $\overline{\alpha}_t := \prod_{t=1}^{N} \alpha_t$, where β_t refers to the noise added at each step t during the forward diffusion process.

Finally, they update the estimate $\boldsymbol{x}_{t-1}^g$ from the unconditional proposal $\hat{\boldsymbol{x}}_{t-1}^g$ by conditioning with the predefined low frequency prior $\Phi_{\text{lp'}}$,

$$\boldsymbol{x}_{t-1}^g = \hat{\boldsymbol{x}}_{t-1}^g - w\nabla_{\boldsymbol{x}_t} \left\| \Phi_{\text{lp'}}(\boldsymbol{x}_0) - \Phi_{\text{lp'}}(\hat{\boldsymbol{x}}_0^g) \right\|_2, \tag{2}$$

where $\Phi_{\text{lp'}}(\cdot)$ is the image-level, predefined low-pass filter, and the scale weight w controls the step size of the guidance. They derive the low-pass filtering effect by downsampling the image followed by upsampling back to the original image dimensions. They then iteratively update the estimate of the denoised image $\boldsymbol{x}_{t-1}^g$ until $t = 0$.

We derive the denoised image $\boldsymbol{x}_{t-1}^g$ from the unconditional proposal $\hat{\boldsymbol{x}}_{t-1}^g$,

$$\begin{aligned}\boldsymbol{x}_{t-1}^g =& \hat{\boldsymbol{x}}_{t-1}^g - w\nabla_{\boldsymbol{x}_t}(\left\|\Phi_{\text{lp}}(\boldsymbol{x}_0) - \Phi_{\text{lp}}(\hat{\boldsymbol{x}}_0^g)\right\|_2 \\ &+ \left\|\Phi_{\text{hp}}(\boldsymbol{x}_0) - \Phi_{\text{hp}}(\hat{\boldsymbol{x}}_0^g)\right\|_2),\end{aligned} \tag{3}$$

where $\Phi_{\text{lp}}(\cdot), \Phi_{\text{hp}}(\cdot)$ are our learned, spatially adaptive low and high frequency priors respectively, and the scale weight w controls the step size of the guidance. We iteratively update the estimate of the denoised image $\boldsymbol{x}_{t-1}^{g}$. The estimate of the denoised image is then passed to the task model for segmentation/depth estimation.

3.2 Y-shaped frequency prediction network for visual filtering

In this section, we present our frequency-based filtering method for robust test-time input adaptation. We process a corrupted image $\mathbf{I}^c \in \mathbb{R}^{H\times W}$ for pixel-wise filtering,

$$\mathbf{I}^f = \sum_{i=1}^{l=2} \mathbf{K}_i \odot \mathbf{I}^c, \tag{4}$$

where $\mathbf{I}^f \in \mathbb{R}^{H\times W}$ is the estimate of the denoised image and $\odot$ denotes the pixel-wise filtering operation. $\mathbf{K}_i \in \mathbb{R}^{H\times W\times K^2}$ denotes the kernels of size $K \times K$ ($K = 3$). We jointly apply $\mathbf{K}_{i=1,2}$ to the corrupted image $\mathbf{I}^c$ to generate $\mathbf{I}^f$.

Encouraged by the successful application of Kernel Prediction Networks (KPN) across several tasks [47–49], we leverage KPNs to learn the spatially adaptive kernels for conditioning diffusion models. We propose to estimate the kernels $\mathbf{K}_{i=1,2}$ from a noisy image $\mathbf{I}^c$,

$$\mathbf{K}_{i=1,2} = \text{Y-FPN}(\mathbf{I}^c), \tag{5}$$

where Y-FPN denotes the Y-shaped Frequency Prediction Network and shares a similar architecture with UNet [50]. Unlike previous work [47–49], which learn a single kernel for each task, Y-FPN predicts 2 separate kernels $\mathbf{K}_{i=1}$ and $\mathbf{K}_{i=2}$ per pixel for a given image.

For a given pixel p in the corrupted image $\mathbf{I}^c$, we derive the low frequency priors Φ_{lp} from $\mathbf{K}_{i=1}$ by applying the method [51], which normalizes $\mathbf{K}_{i=1}$ with a spatial softmax:

$$\mathbf{K}_{i=1}[u,v] = \frac{\exp(\mathbf{K}_{i=1}[u,v])}{\sum_{u',v'} \exp(\mathbf{K}_{i=1}[u',v'])}, \tag{6}$$

where (u, v) indexes the spatial location in the $K \times K$ kernel. We then compute the low-frequency prior for pixel p

$$\Phi_{\text{lp}}(\mathbf{I}^c)[p] = \sum_{u,v} \mathbf{K}_{i=1}[u,v].\mathbf{I}^c[p]. \tag{7}$$

We also normalize $\mathbf{K}_{i=2}$ with a spatial softmax (Eq. 6) and compute the following equation to obtain the high frequency prior for pixel p

$$\Phi_{\text{hp}}[p](\mathbf{I}^c) = \mathbf{I}^c - \sum_{u,v} \mathbf{K}_{i=2}[u,v].\mathbf{I}^c[p]. \tag{8}$$

To train Y-FPN, we apply the commonly used loss functions for image restoration *i.e.* L_1, SSIM (Structural Similarity) and a Frequency Reconstruction Loss [52].

3.3 Loss functions for training Y-FPN

We apply the commonly used loss functions for image restoration *i.e.* L_1, SSIM (Structural Similarity) to train our network. We also include a Frequency Reconstruction Loss [52] $\mathcal{L}_{Freq}$. We first apply the Fourier Transform to the filtered image $\mathbf{I}^f$ and the original clean image $\mathbf{I}$. We then compute the difference between Fourier transformed images and normalize the result with a logarithmic function

$$\mathcal{L}_{Freq}(\mathbf{I}^f, \mathbf{I}) = \log(1 + \frac{1}{HW}\|\mathcal{F}(\mathbf{I}) - \mathcal{F}(\mathbf{I}^f)\|), \tag{9}$$

where H, W are the spatial dimensions of the image in frequency space. Bringing it all together, we have the following overall training loss for Y-FPN,

$$\mathcal{L}(\mathbf{I}^f, \mathbf{I}) = \|\mathbf{I}^f - \mathbf{I}\|_1 - \lambda_1 \text{SSIM}(\mathbf{I}^f, \mathbf{I}) + \lambda_2 \mathcal{L}_{Freq}(\mathbf{I}^f, \mathbf{I}). \tag{10}$$

where λ_1, λ_2 are hyperparameters. We fix λ_1, λ_2=0.2 for all experiments.

3.4 Loss functions for test-time adaptation

Here, we provide a brief description of the loss functions during test-time adaptation. For semantic segmentation, we provide additional supervision from the FOCUS denoised images via the standard cross entropy loss,

$$\mathcal{L}_{FOCUS}(G, \boldsymbol{X}, \boldsymbol{Y}') = \sum_{i=1}^{H\times W} \sum_{c=1}^{C} -Y'_{ic} log(G(\boldsymbol{X})), \tag{11}$$

where $G(\boldsymbol{X})$ refers to the predicted probability of class c for the ith pixel for the input image $\boldsymbol{X}$. Y'_{ic} is the predicted label from the task network G on the FOCUS denoised images for class c on the ith pixel, where $Y'_{ic} = 1$ if the pixel belongs to the class c and $Y'_{ic} = 0$ if otherwise.

In preliminary experiments for depth estimation on NYU2k-C, we used the standard SILog loss [53] to update the baseline model adaptation methods. However, we observed that the model adaptation methods converged to degenerate solutions (e.g., predicting constant depth), which could be caused by unstable gradients in the presence of strong corruptions.

To enable a fair comparison and ensure convergence, we use the L1 loss, which yielded more stable results. We provide a description below on the loss functions used for depth estimation. Given an input $\boldsymbol{x}$ and its horizontally flipped image $\boldsymbol{x}_{flip}$, we compute the depth using the depth estimation model D.

$$\boldsymbol{p} = D(\boldsymbol{x}), \tag{12}$$

$$\boldsymbol{p_{flip}} = D(\boldsymbol{x}_{flip}), \tag{13}$$

We derive the continuous-valued confidence mask s as follows

$$s = \frac{1}{|\boldsymbol{p_{flip}} - \boldsymbol{p}| + 10^{-6}}. \tag{14}$$

We then optimize the model adaptation methods using the following consistency loss

$$\mathcal{L}(D, \boldsymbol{x}, \boldsymbol{x}_{flip}) = |\boldsymbol{p_{flip}} - \boldsymbol{p}| + s|\boldsymbol{p} - D(\boldsymbol{x})|. \tag{15}$$

For FOCUS-driven supervision, we include the FOCUS denoised images $\boldsymbol{x}_{FOCUS}$,

$$\mathcal{L}_{FOCUS}(D, \boldsymbol{x}, \boldsymbol{x}_{FOCUS}) = s|\boldsymbol{p} - D(\boldsymbol{x}_{FOCUS})|. \tag{16}$$

3.5 FrequencyMix training

In this section, we introduce our approach to generate sufficiently challenging data for training the Y-shaped Frequency Prediction Network (Y-FPN). Since Y-FPN must be robust to noise across a broad range of frequencies, we apply frequency-dependent perturbations and combine these frequency-dependent perturbations via FrequencyMix to increase the training data diversity.

3.5.1 Frequency dependent perturbations

Inspired by the success of previous work [40] that improved model generalizability by perturbing the amplitude spectrum of the high spatial frequency region, we adapted their approach to include additional perturbations.

For a single-channel image $\boldsymbol{x} \in \mathbb{R}^{H \times W}$, the Fourier transform $\mathcal{F}(\boldsymbol{x})$ is described as

$$\mathcal{F}(\boldsymbol{x})[m, n] = \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} \boldsymbol{x}[h, w] \exp\left(-2\pi i(\frac{h}{H}m + \frac{w}{W}n)\right), \tag{17}$$

where $i^2 = -1$ and m, n refer to the spatial frequencies in Fourier space.

We can then derive the amplitude spectrum $\mathcal{A}(\boldsymbol{x})$ and the phase spectrum $\mathcal{P}(\boldsymbol{x})$ as follows

$$\mathcal{A}(\boldsymbol{x}) = \sqrt{(\Re\{\mathcal{F}(\boldsymbol{x})[m, n]\})^2 + \Im\{\mathcal{F}(\boldsymbol{x})[m, n]\})^2)}, \tag{18}$$

$$\mathcal{P}(\boldsymbol{x}) = arctan\left(\frac{\Im\{\mathcal{F}(\boldsymbol{x})[m, n]\}}{\Re\{\mathcal{F}(\boldsymbol{x})[m, n]\}}\right), \tag{19}$$

where $\Re\{\mathcal{F}(\boldsymbol{x})[m, n]\}$ and $\Im\{\mathcal{F}(\boldsymbol{x})[m, n]\}$ denote the real and imaginary components of the Fourier Transform respectively. Since we are working with RGB images, we compute the Fourier Transform for each channel independently.

To perturb the clean images, we introduce $\epsilon_{high}[m, n]$, $\epsilon_{low}[m, n]$, $\epsilon_{rand}[m, n]$ and ϵ_{uni}. We use the spatially dependent functions to control the perturbation extent.

$$\epsilon_{high}[m, n] \sim \mathcal{N}\left(1, \left(2\alpha\sqrt{\frac{m^2 + n^2}{H^2 + W^2}} + \beta\right)^2\right), \tag{20}$$

$$\epsilon_{low}[m, n] \sim \beta\mathcal{N}\left(1, 2e^{-\alpha(m^2+n^2)}\right), \tag{21}$$

$$\epsilon_{rand}[m, n] \sim \mathcal{N}\left(1, \alpha\frac{(H - m + y_{pk})^2 + (W - n + x_{pk})^2}{H^2 + {y_{pk}}^2 + W^2 + {x_{pk}}^2}\right) \tag{22}$$

$$\epsilon_{uni} = \alpha\beta, \tag{23}$$

where severity $\alpha \sim (3, 4, 5)$, $x_{pk} \sim Uniform(0, W/4)$, $y_{pk} \sim Uniform(0, H/4)$, $\beta = 0.3$,

$m \in [-H/2, H/2]$, $n \in [-W/2, W/2]$. We zero-centre the amplitude spectrum $\mathcal{A}(\boldsymbol{x})$ before applying the perturbation function $g(.)$

$$\begin{aligned}\hat{\mathcal{A}}(\boldsymbol{x}) &= g(\mathcal{A}(\boldsymbol{x}), \epsilon)[m, n] \\ &= (1 + \epsilon\mathcal{A}(\boldsymbol{x})),\end{aligned} \tag{24}$$

where $\epsilon \in [\epsilon_{rand}, \epsilon_{high}, \epsilon_{low}, \epsilon_{uni}]$. Finally, we obtain the augmented image via 2D-FFT by recombining the perturbed amplitude spectrum $\hat{\mathcal{A}}(\boldsymbol{x})$ with the original phase spectrum $\mathcal{P}(\boldsymbol{x})$.

3.5.2 FrequencyMix

While we have addressed the difficulty of perturbing images as a function of spatial frequency, we are still faced with the challenge of generating sufficiently diverse data for training Y-FPN. Previous work demonstrated the effectiveness of data augmentation by randomly sampling and combining

data transformation operations for robust image classification [54] and image de-raining [49]. Similar to Augmix [54], we randomly sample and combine the frequency-dependent perturbations: low-frequency, high-frequency and randpeak (randomized frequency peaks).

- ADE20k [56, 57] is a densely annotated semantic segmentation dataset of indoor and outdoor images. We use 25,574 images for training Y-FPN and 2000 images for evaluation. Since there is randomness involved during the corrupted image generation, we use the publicly available ADE20k-C validation set from Li *et al.*[6] to preserve reproducibility.
- Cityscapes [58] is a real-world driving semantic segmentation dataset with densely annotated images of resolution 2048 × 1024 pixels. We train Y-FPN with the default split of 2975 images. We apply the corruptions from ImageNet-C on the evaluation split of 500 images.
- NYU2k [59] is a densely annotated dataset of indoor scenes for depth estimation. We train Y-FPN with the standard split of 50,688 images and evaluate performance on the test split of 654 images at the original resolution of 640 × 480 pixels after applying the corruptions from ImageNet-C.

Algorithm 1 FrequencyMix and Y-FPN training

Input: Clean Images $\mathcal{I}$, Y-like Frequency Prediction Network Y-FPN(.),
Loss function $\mathcal{L}$., Augmentations $Aug = [\text{Uni. Noise}, g(\epsilon_{low}), g(\epsilon_{high}), g(\epsilon_{uni}), g(\epsilon_{rand})]$
Output: Trained Y-like Frequency Prediction Network Y-FPN(.)
Function FrequencyMix(clean Image $\mathbf{I}$) :
1: Sample mixing weights $(w_1, w_2, w_3, w_4) \sim Dirichlet$; [1]
2: Initialize an empty image $\mathbf{I}_{mix}$;
3: **for** $i = 1, ..., 4$ **do**
4: Sample augmentations $(a_1, a_2, a_3, a_4) \sim Aug$;
5: Combine augmentations via $a_{12} = a_2 a_1$, $a_{123} = a_3 a_2 a_1$ and $a_{1234} = a_4 a_3 a_2 a_1$;
6: Sample operation $o \sim (a_1, a_{12}, a_{123}, a_{1234})$;
7: $\mathbf{I}_{mix} += w_i o(\mathbf{I})$;
8: **end for**
9: Sample a blending weight $w \sim Beta$;
10: **return** $\mathbf{I}_{noisy} = w\mathbf{I} + (1 - w)\mathbf{I}_{mix}$
11: **for** $i = 0, ..., iter_{num}$ **do**
12: Sample a clean Image $\mathbf{I} \sim \mathcal{I}$;
13: Generate noisy image with FrequencyMix $\mathbf{I}_{noisy} \leftarrow$ FrequencyMix($\mathbf{I}$);
14: Generate estimate of clean image with Y-FPN $\hat{\mathbf{I}} \leftarrow$ Y-FPN($\mathbf{I}_{noisy}$);
15: Calculate Loss $\mathcal{L}$ from eqn 10 and backpropagate ;
16: Update Y-FPN(.) parameters.
17: **end for**

[1] We randomly sample the augmentation depth at each iteration from the set {1,2,3,4}. The augmentation depth determines the number of mixing weights, the number of iterations in line 3 and the number of sampled augmentations in line 4. For this case, the augmentation depth is 4.

4 Experimental details

In this section, we provide an overview of the datasets (Sect. 4.1), baselines (Sect. 4.2) and implementation details (Sect. 4.3).

4.1 Datasets

Here, we describe the datasets used in this work.

- ImageNet-C [55] We apply the corruptions drawn from the categories (*i.e. Noise*, *Blur*, *Digital*, *Weather*) in ImageNet-C [55] at the highest severity (level=5) to the evaluation images.

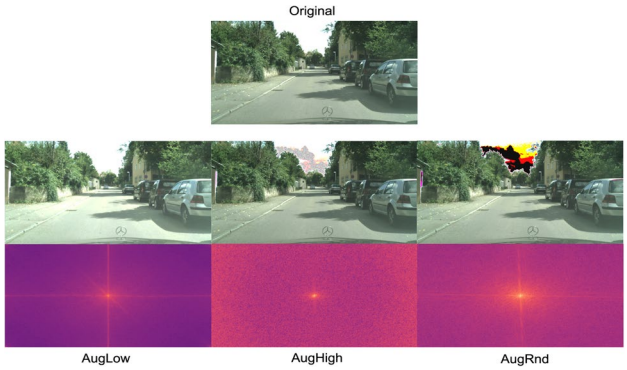

Fig. 3 Examples of amplitude perturbed training images 'AugLow', 'AugHigh' and 'AugRnd'. Top Row: Augmented Images 'AugLow', 'AugHigh' and 'AugRnd'. Bottom Row: FFT of the difference images (Augmented Image-Original)

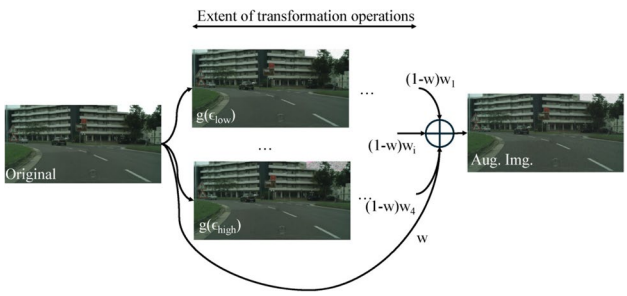

Fig. 4 Illustration of FrequencyMix. We combine randomly selected augmentations and re-weight the augmented images to generate a composite augmented image. The weights w_i are sampled from a Dirichlet distribution, and the blending weight w is sampled from a Beta distribution

4.2 Baselines

- TENT [4] is a keystone paper for test-time adaptation, having demonstrated effectiveness for image classification and semantic segmentation. It updates the affine parameters within the batch normalization layers via entropy minimization. Since entropy is not well-defined for depth estimation, we apply a consistency loss between original and augmented predictions.
- CoTTA [5] is a model adaptation method that utilizes a exponential mean weighting average model as a teacher to generate reliable pseudo labels for test time adaptation. CoTTA addresses catastrophic forgetting by stochastically updating the weights from the teacher model with the weights from the pretrained source model.
- DDA [7] utilizes a predefined image-level low-pass filter to condition a pretrained ImageNet diffusion model for denoising of corrupted images in pixel-space. Unlike model adaptation methods that are sensitive to batch sizes and learning rates, DDA is invariant to these test-time hyperparameters.
- DUSA [6] is a recent state-of-the-art work utilising a latent space diffusion model for test-time adaptation. Leveraging a ControlNet trained on ADE20k, they use the prediction logits to condition the diffusion model. They then estimate the noise at a single timestep (N=100) and optimize the task model and the diffusion model by minimizing the error between the estimated noise and the actual noise. We use their publicly available code for comparison.

4.3 Implementation details

We use the publicly available pretrained models (Segformer-b5 [60] for semantic segmentation, and DenseDepth [61] for depth estimation). For DDA, we use the same number of timesteps and scaling weight as FOCUS. For Y-FPN training, we use learning rate 2×10^{-4}, Adam optimizer with $\beta_1 = 0.5, \beta_2 = 0.999$. We describe FrequencyMix and Y-FPN training in Algorithm 1 and show the FOCUS workflow in Algorithm 2 (Figs. 3 and 4).

Semantic segmentation For semantic segmentation tasks, we use the implementation for TENT, CoTTA and DUSA from Li *et al.*[6]. We perform model adaptation (TENT, CoTTA, DUSA) with a batch size of 1, Adam optimizer with a learning rate of $6 \times 10^{-5}/8$. For Cityscapes, we use the hyperparameters $\lambda_1 = 0.2, \lambda_2 = 0.2$ to train Y-FPN. During diffusion-driven denoising, we use scale weights $w = 2$ and the number of timesteps $nsteps = 15$.

For ADE20k, we train Y-FPN with $\lambda_1 = 0.2, \lambda_2 = 0.2$. During diffusion-driven denoising, we use scale weights $w = 2$ and the number of timesteps, $nsteps = 25$.

Depth estimation For the model adaptation methods (TENT, CoTTA), we use a batch size of 1, Adam optimizer. We evaluate performance at learning rate 1×10^{-4} and 1×10^{-5}. we train Y-FPN with $\lambda_1 = 0.2, \lambda_2 = 0.2$. During diffusion-driven denoising, we use scale weights $w = 3$, and the number of timesteps, $nsteps = 25$.

Algorithm 2 **F**requency **O**ptimized **C**onditioning for Diff**US**ion.

Input: Noisy Image $\boldsymbol{I_{noisy}}$, Trained Y-like Frequency Prediction Network Y-FPN(.), Pretrained (ImageNet) Denoising Probabilistic Diffusion Model, Number of Timesteps n

Output: Denoised image $\boldsymbol{x_0^g}$

1: Obtain estimate of denoised image $\boldsymbol{x_0} \leftarrow$ Y-FPN($\boldsymbol{I_{noisy}}$) ;
2: Sample $\boldsymbol{x}_N \sim q(\boldsymbol{x}_N|\boldsymbol{x}_0)$
3: **for** $t = 1000...1000 - n$ **do**
4: $\hat{\boldsymbol{x}}^g_{t-1} \sim p_\theta(\boldsymbol{x}^g_{t-1}|\boldsymbol{x}^g_t)$;
5: $\boldsymbol{x}^g_{t-1} \leftarrow \hat{\boldsymbol{x}}^g_{t-1} - \boldsymbol{w}\nabla_{\boldsymbol{x}_t} \left\|\Phi_{\text{lp}}(\boldsymbol{x}_0) - \Phi_{\text{lp}}(\hat{\boldsymbol{x}}^g_0)\right\|_2 + \left\|\Phi_{\text{hp}}(\boldsymbol{x}_0) - \Phi_{\text{hp}}(\hat{\boldsymbol{x}}^g_0)\right\|_2$.
6: **end for**

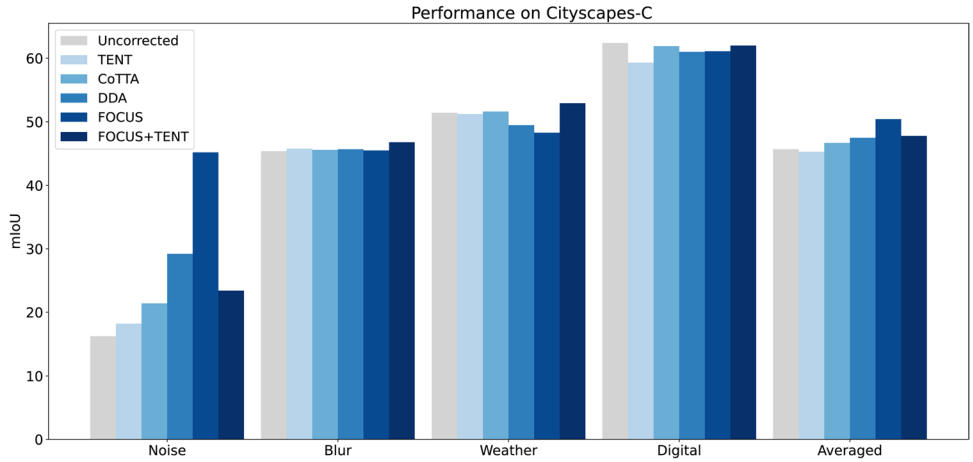

Fig. 5 Quantitative results for Cityscapes-C. FOCUS demonstrates the strongest performance ('Averaged') and demonstrates strong synergy ('FOCUS+TENT') for the corruption categories 'Blur', 'Weather' and 'Digital'. Note that 'FOCUS+TENT' demonstrates stronger averaged performance compared to 'TENT' across all 4 corruption categories. Full per-corruption results in Table 13

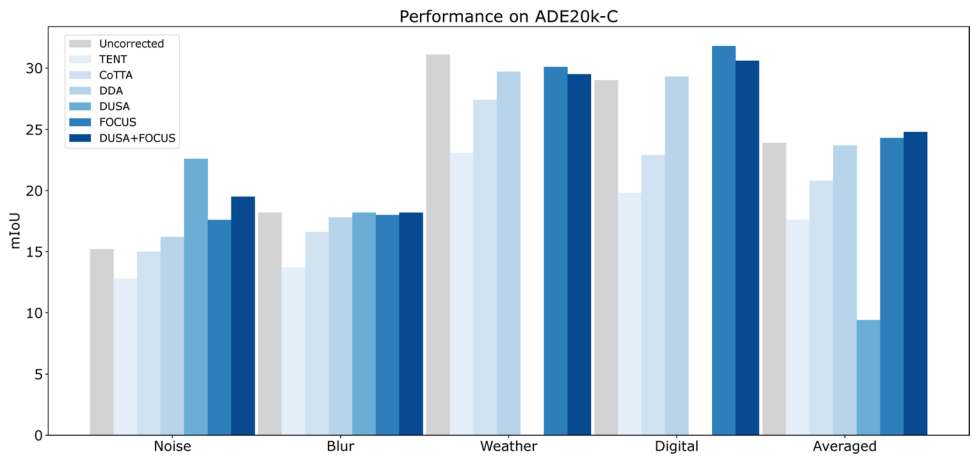

Fig. 6 Quantitative results for ADE20k-C. Performance for DUSA [6] starts strong initially, but degrades rapidly from catastrophic forgetting ('Weather', 'Digital'). FOCUS also demonstrates stronger performance compared to other model adaptation (*e.g.*TENT, CoTTA) and input adaptation methods (DDA). Additionally, 'FOCUS+DUSA' recovers the lost performance and even surpasses 'FOCUS', suggesting the potential benefits of combining model adaptation methods with input adaptation methods

4.4 Metrics

We adopt the commonly used mean Intersection over Union(mIoU) and the Averaged Absolute Relative Error (Avg. Abs. Rel. Error) as the quantitative evaluation metrics for semantic segmentation and depth estimation respectively. A higher mIoU and a lower Avg. Abs. Rel. Error indicates stronger performance.

4.5 Test-time adaptation performance

Semantic segmentation Our method outperforms the baselines on averaged mIoU for Cityscapes-C (Fig. 5) and ADE20k-C (Fig. 6). On average, FOCUS performs better than DDA, which uses predefined, image-level low frequency priors, indicates the importance of learned spatially adaptive priors. Across Cityscapes and ADE20k, we observe stronger performance compared to DDA across nearly all corruption types except for 'Weather' (Cityscapes-C).

While DUSA [6] performs well initially, performance decreases rapidly after model fine tuning on a few corruptions. We suggest that the reduced performance can be attributed to the following causes. Firstly, the task model may overfit to test data, reducing the quality of the prediction logits when the test data distribution shifts significantly. Since the diffusion model is conditioned on the prediction logits, the reduced quality of the prediction logits worsens the noise estimates, which sends erroneous supervisory signals to the task model and the diffusion model. Additionally, since the diffusion model is also finetuned on the test data, some of the existing knowledge may be degraded as the diffusion model overfits to the test data. These factors might account for the steep performance drop compared to CoTTA and TENT.

We also show some qualitative results in Figs. 8 and 9. Importantly, model adaptation methods are heavily influenced by original task model performance ('Uncorrected'). For example, the source pretrained depth estimation model ('Uncorrected') performs poorly on the image degraded by 'Contrast'. Consequently, both TENT and CoTTA perform poorly, underscoring the key requirement that strong out-of-distribution performance of the task model is essential for effective test time adaptation [62], In contrast, FOCUS is not as reliant on task model performance and provides additional improvements when combined with model adaptation methods ('FOCUS+TENT').

Table 1 Computational cost comparison of adaptation methods for ADE20k-C

Method	Trainable params (M)	Time (s/frame)	Memory (GB)
Baseline	0	0.12	∼ 2
TENT	0.4	0.18	∼ 6
CoTTA	81.4	2.64	∼ 12
DDA	0	15.58	∼ 24
DUSA	1300	4.08	∼ 65
FOCUS	27.5	17.72	∼ 24
FOCUS+DUSA	109	19.21	∼44*

*We add up the GPU memory costs from diffusion-driven denoising and fine tuning the task model

Table 2 Quantitative results (averaged absolute relative error ↓) for depth estimation (NYU2k-C). **Bold** denotes best results

Method	LR = 10^{-4}	LR = 10^{-5}
Uncorrected	0.710	0.710
TENT	0.625	0.378
CoTTA	*inf*[3]	0.321
DDA	0.80	0.80
FOCUS	0.479	0.479
FOCUS + TENT	**0.418**	0.361
FOCUS + CoTTA	*inf*[2]	**0.269**

[2]Relative error diverged due to degenerate model output

Deblurring For deblurring, FOCUS generally performs better than the other baseline methods (Fig. 9). Generally, we observe reduced gains on 'Blur' and 'Weather' categories. These corruptions cause signal loss via occlusions ('Frost','Fog'), and degrade structural information via blurring. For model adaptation methods (*e.g.*TENT, CoTTA), performance is lower than the baseline 'Uncorrected' for ADE20k-C (Fig. 6) and comparable with the baselines 'Uncorrected' for Cityscapes-C (Fig. 5). We suggest the reduced performance is caused by poorer pseudo label quality resulting from model overfitting to the earlier corruption category 'Noise'.

While specialized models [29, 63] tailored to address specific corruption categories such as blur can improve performance for visually similar corruptions, these methods are generally sensitive to high frequency noise as the sharpening process may worsen the severity of the image corruptions from the high frequency noise.

Recent work [64] introduce a noise-aware classifier to characterize the corruptions in the image, enabling the use of corruption-specific denoising techniques. However, training such a classifier requires prior knowledge and is complex to implement.

Synergy with model adaptation In their comprehensive review of image restoration, Zhang *et al.*[65] observe that unsupervised methods may provide a potential solution to dealing with this problem. Following along this line of thought, we present a simple, straightforward solution to address these 'antagonistic corruptions'. The FOCUS denoised images provide an additional source of supervision for model adaptation methods [4, 6] via the standard cross-entropy loss for semantic segmentation and consistency loss for depth estimation. We describe the additional loss functions in Sect. 3.4.

Integrating input-adaptation methods such as FOCUS with model adaptation methods yields synergistic effects for both depth estimation and semantic segmentation. We note that FOCUS mitigates catastrophic forgetting for model adaptation-based methods such as TENT [4] and DUSA [6]. FOCUS improves adaptation performance for semantic segmentation (Figs. 5 and 6) and for depth estimation (Table 2), highlighting the synergistic effects of combining input and model adaptation.

Depth estimation For depth estimation, FOCUS outperforms the benchmarks at learning rate 10^{-4} and demonstrates competitive performance at learning rate 10^{-5} in Table 2. While TENT, CoTTA are sensitive to learning rate changes, we show that FOCUS can stabilize performance for TENT ('FOCUS+TENT') across learning rates.

TENT only updates the affine parameters instead of updating all parameters (CoTTA). This might make TENT less sensitive to changes in learning rate. Nevertheless, FOCUS improves performance for both TENT and CoTTA

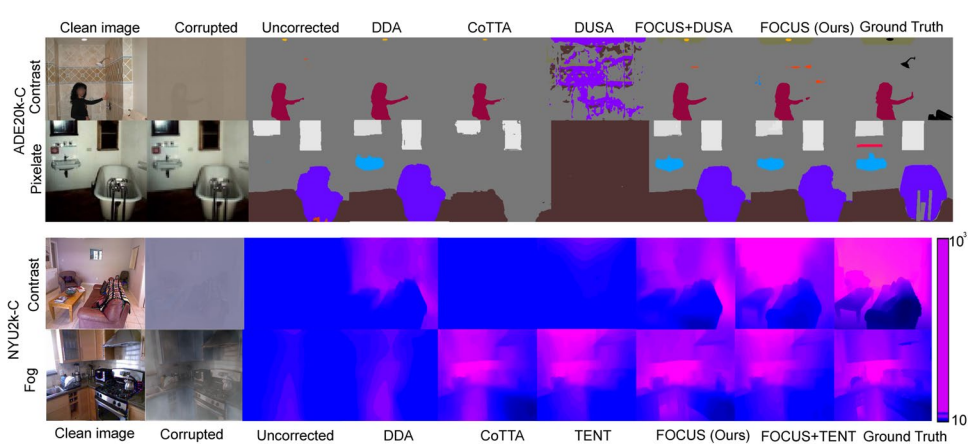

Fig. 8 Qualitative examples for semantic segmentation (ADE20k-C) and depth estimation (NYU2k-C). Full per-corruption results in Tables 14, 15, and 16

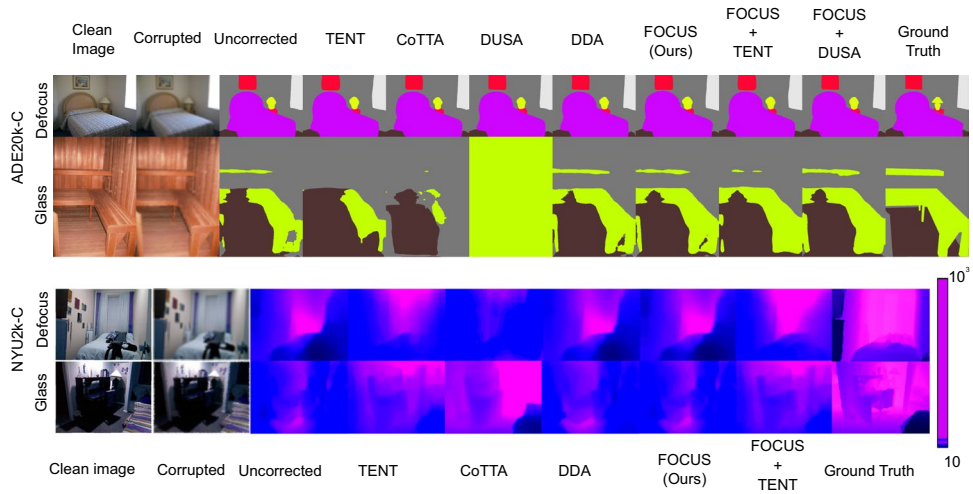

Fig. 9 Qualitative examples of deblurring for semantic segmentation (ADE20k-C) and depth estimation (NYU2k-C)

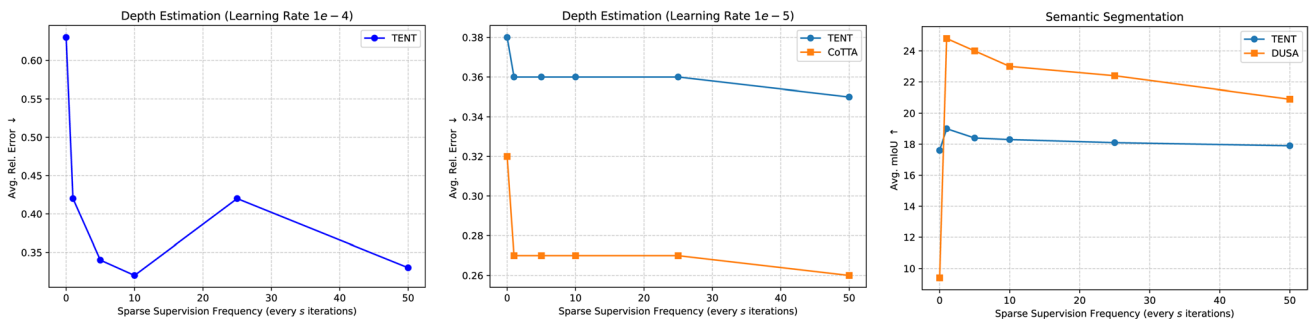

Fig. 7 Performance for sparse supervision using pseudo labels from FOCUS denoised images on a) Depth Estimation (NYU2k-C, Learning Rate $1e-4$) b) Depth Estimation (NYU2k-C, Learning Rate $1e-5$) c) Semantic Segmentation (ADE20k-C). We provide additional supervision every s iterations - /*i.e.* larger i$s\rightarrow$ more sparse supervision

(learning rate 10^{-4}), highlighting the utility of combining input and model adaptation.

4.6 Computational costs

Training Y-FPN requires around 15GB GPU memory. For FOCUS, we use the ImageNet pretrained diffusion model without any additional finetuning. Running the diffusion-driven diffusion denoising requires around 32GB of GPU memory (NVIDIA L40 46GB) for a batch size of 1 for an input image resolution 1024×512 pixels (Cityscapes-C), around 24GB for 512×512 pixels (ADE20k-C) and about 28GB for 640×512 pixels (NYU2k-C).

Our work attempts to strike a balance between performance and minimizing computational costs. The computational costs at test time in terms of GPU memory are higher than conventional test time adaptation methods (TENT, CoTTA), but less than those of DUSA (Table 1). Since we do not fine-tune the diffusion model for FOCUS+DUSA, we only require around 20GB GPU memory for model adaptation. Combined with the GPU memory required during FOCUS diffusion-driven denoising, we use a total of approximately 44GB of GPU memory, while DUSA requires around 65GB of GPU memory at test time.

However, since inference speeds for diffusion-driven methods pose a bottleneck, we study the effect of intermittent supervision on adaptation performance in the following section (Sect. 4.7).

4.7 Frequency of FOCUS-driven supervision for model adaptation

To investigate the balance between adaptation effectiveness and inference efficiency, we studied the effect of intermittent supervision from the FOCUS denoised images on other model adaptation methods for depth estimation and semantic segmentation (Fig. 7). Specifically, we update the task model for the baselines (semantic segmentation on ADE20k-C: TENT [4], DUSA [6] and depth estimation on NYU2k-C: TENT [4], CoTTA [5]) with additional supervision from the FOCUS denoised images every (1, 5, 10, 25, 50) iterations. This design evaluates whether FOCUS can still regularize model adaptation despite limited supervision.

Our results on ADE20k-C reveal that while performance decreases as the frequency of supervision from FOCUS

Table 3 Evaluation of λ_1 during Y-FPN training for semantic segmentation (Cityscapes-C)

λ_1	Gaussian Noise	Motion Blur	Snow	Pixelate
0	2.97	53.93	73.03	76.17
0.1	43.07	55.18	69.12	73.61
0.2	41.97	55.17	67.10	74.10

We use mIoU (↑) for the performance metrics

Table 4 Evaluation of frequency loss weights during Y-FPN training for semantic segmentation (Cityscapes-C). **Bold** denotes best results

λ_2	Gaussian Noise	Defocus Blur	Snow	Pixelate
0.01	41.40	**55.80**	29.73	**74.49**
0.1	32.99	54.24	23.02	73.77
0.2	**41.97**	55.08	**33.49**	74.10
0.4	26.26	55.29	24.93	73.89

We use mIoU (↑) for the performance metrics

Table 5 Comparison of scale weights for semantic segmentation (Cityscapes-C). **Bold** denotes best results

Scale weight	Gaussian Noise	Defocus Blur	Snow	Pixelate
0.5	39.03	53.20	29.12	71.71
1	40.70	54.56	31.26	73.19
2	41.97	**55.08**	**33.49**	**74.10**
4	**42.33**	54.76	30.03	73.80

We use mIoU (↑) for the performance metrics

Table 6 Comparison of scale weights for depth estimation (NYU2k-C). **Bold** denotes best results

Scale weight	Gaussian Noise	Defocus Blur	Snow	Pixelate
1	0.34	1.40	0.75	0.40
3	**0.22**	1.24	**0.69**	0.25
6	0.22	**1.07**	0.71	**0.16**

We use Averaged Absolute Relative Error (↓) for the performance metrics

decreases, performance is still higher compared to model adaptation performance without supervision (k=0). We found that adaptation performance at lower supervision frequencies ($k = 5, 10$) is comparable with supervision at $k = 1$ for both semantic segmentation and depth estimation tasks, suggesting the effectiveness of regularising model adaptation with pseudo labels from denoised images. Additionally, performance improved at $k = 50$ (supervision from FOCUS denoised images is applied every 50 iterations) relative to $k = 0$, demonstrating the effectiveness of sparse FOCUS-driven supervision for model adaptation methods.

For depth estimation, we observe that additional supervision with FOCUS reduces the impact of increasing learning rate for model adaptation on depth estimation tasks. Interestingly, we note for TENT and CoTTA, varying the frequency of FOCUS-driven supervision at learning rate 10^{-5} does not negatively impact performance. Furthermore, at higher learning rates 10^{-4}, we observe that reducing the frequency of FOCUS-driven supervision can even improve performance. This suggests that applying denoised images, even intermittently and rarely, have a strong regularization effect for test-time model adaptation methods.

Table 7 Ablation study on ADE20k-C (semantic segmentation) and NYU2k-C (depth estimation)

Method	ADE20k-C (mIoU) ↑	NYU2k-C (Abs. Rel. Err.) ↓
Baseline	23.9	0.710
+ Y-FPN only	22.3	0.646
+ Unconditioned Diff	22.2	0.669
+ Low-Freq. Prior	23.8	0.599
+ High-Freq. Prior	24.2	0.605

Metrics: mIoU (↑) and Abs. Rel. Error (↓)

4.8 Effect of the loss weight λ_1

In this section, we evaluate the effect of λ_1 on adaptation performance (Table 3). We conducted additional experiments on Cityscapes-C. We note that removing the SSIM loss term significantly degrades performance for 'Gaussian Noise', suggesting the importance of SSIM in preserving local structure. Performance for 'Motion Blur' is reduced by a slightly smaller degree when the SSIM loss term is removed, which indicates the need to preserve local correlations for handling blur type corruptions. Performance is improved for 'Snow' and 'Pixelate' when Y-FPN is trained without the SSIM loss term, which suggest that SSIM is less effective for dealing corruptions that follow regular patterns ('Pixelate') or introduced localized irregularities ('Snow').

4.9 Effect of the frequency loss weight λ_2

In this section, we evaluate the effect of λ_2 on adaptation performance (Table 4). We observe that λ_2 affects adaptation performance for 'Noise' ('Gaussian Noise'), followed by 'Weather' ('Snow') and 'Blur'('Defocus Blur') type corruptions. We found that optimal hyperparameter values lie within the range [0.1,0.2].

4.10 Scale weight *w*

Scale weight is another important hyperparameter because it directly controls the conditioning strength of the frequency priors during diffusion-driven denoising. We found that increasing the scale weights generally improved performance for semantic segmentation (Table 5) and depth estimation (Table 6), which aligns with our expectation that preserving the semantic information during denoising improved adaptation performance.

4.11 Ablation study

We present ablation studies for ADE20k-C and NYU2k-C (Table 7).

We find that FOCUS achieves noticeable improvements compared to FOCUS (Y-FPN only), indicating that conditioned diffusion-driven denoising improves performance. Unconditioned diffusion-driven denoising, however, can degrade performance, further highlighting the importance of conditioning diffusion model during image denoising to preserve semantic information. **Comparison of Frequency Priors** Conditioning with learned low frequency priors only improves performance for both semantic segmentation and depth estimation tasks. Including high frequency priors further improves performance for semantic segmentation, but slightly reduces it for depth estimation. This suggests that conditioning with low frequency priors is generally sufficient for good performance. However, for corruptions degrade high frequency information (e.g. 'Contrast') we observed that conditioning with high and low frequency priors improves performance. To further investigate the effect of frequency perturbations on model adaptability, we perform the comparison study in the following section (Sect. 4.11.1).

4.11.1 Effect of different frequency perturbations in FrequencyMix

Since training data diversity is an important factor determining generalizability, we evaluate the effect of the 3 main frequency perturbations (Table 8). We train 3 Y-FPN network variants by combining the base augmentations (uniform noise and uniform amplitude perturbations) with one of the 3 main frequency perturbations (AugLow, AugHigh and AugRnd) on Cityscapes. We observe that AugHigh is more effective than AugLow or AugRnd in improving performance on 'Gaussian Noise', 'Defocus Blur' and 'Snow'. While AugHigh has the strongest effect on performance compared to AugLow and AugRnd, performance is still lower than Y-FPN trained with all the frequency perturbations. In almost all corruptions, combining all frequency perturbations enhances performance beyond that of each of the frequency perturbations, suggesting the importance of including all the frequency perturbations during Y-FPN training (Table 9).

4.12 Augmentation depth in FrequencyMix

The augmentation depth controls the maximum number of augmentation operations applied to the image at each iteration. Full per-corruption results are provided in Table 17. We randomly select the value for the augmentation depth

Table 8 Evaluation of frequency perturbations during Y-FPN training for FOCUS semantic segmentation (Cityscapes-C). **Bold** denotes best results

	Gaussian Noise	Defocus Blur	Snow	Pixelate
AugLow	28.8	54.4	25.0	**75.2**
AugHigh	40.3	**55.3**	31.5	74.4
AugRnd	31.4	54.7	22.0	72.2
All perturbations	**42.0**	55.1	**33.5**	74.1

We use mIoU (↑) for the performance metrics

Table 9 Evaluation of augmentation depth during Y-FPN training for semantic segmentation (Cityscapes-C). **Bold** denotes best results

Depth *i*	Gaussian Noise	Motion Blur	Snow	Pixelate
{1,2,3,4}	41.97	**55.17**	33.49	74.10
2	35.45	54.13	31.38	**75.82**
4	**46.30**	54.44	**35.09**	73.03
8	44.57	53.14	34.55	74.34

We use mIoU (↑) for the performance metrics. The first row where 1, 2, 3, 4, indicates that the depth is randomly drawn from the set at each iteration

Table 10 Comparison of timesteps for semantic segmentation (Cityscapes-C). **Bold** denotes best results

Timesteps	Gaussian Noise	Defocus Blur	Snow	Pixelate
1	40.6	56.1	32.3	**74.6**
5	40.9	**56.1**	33.2	74.3
15	**42.0**	55.1	**33.5**	74.1
25	40.9	55.1	33.2	74.3
50	41.9	55.2	32.1	73.9

We use mIoU (↑) for the performance metrics

Table 11 Comparison of timesteps for depth estimation (NYU2k-C)

Timesteps	Gaussian Noise	Defocus Blur	Snow	Pixelate
1	1.16	1.26	0.82	0.15
5	1.27	1.15	0.81	0.15
15	1.06	1.05	0.74	0.15
25	0.22	1.24	0.69	0.25
50	0.32	1.06	0.63	0.17

We use Averaged Absolute Relative Error (↓) for the performance metrics

Table 12 Performance on clean images for semantic segmentation (ADE20k-C) and depth estimation (NYU2k-C)

Method	ADE20k-C (mIoU) ↑	NYU2k-C (Avg. Abs. Rel. Error) ↓
Uncorrected	49.1	0.162
TENT	34.7	0.232
CoTTA	46.7	0.174
DDA	48.0	0.179
DUSA	0.0	–
FOCUS	48.0	0.176

Metrics: mIoU (↑) and Avg. Abs. Rel. Error (↓)

from the set {1,2,3,4}. We observe that adaptation performance generally benefits from larger augmentation depths, we note that performance declines slightly at high augmentation depths (e.g. depth= 8). Too much data augmentation might have degraded the original signal in the image, causing the model to focus on removing noise (which may not be representative of noise at test time) instead of learning to extract the semantic information from the distorted images.

4.13 Number of timesteps *N*

For diffusion-driven denoising, timestep selection is important and also widely recognized as a critical hyperparameter for diffusion models. We compare performance for semantic segmentation on Cityscapes-C (Table 10) and depth estimation on NYU2k-C. Our results for segmentation shows that increasing the number of reverse denoising timesteps reduces performance on 'Blur' (*i.e.*'Defocus Blur') but improves it for 'Noise' ('Gaussian Noise'). The results show that the optimal number of time steps are within the range [15,25]. As seen from the variation across timesteps, the number of timesteps affects performance more noticeably for 'Gaussian Noise','Defocus Blur' and 'Snow' compared to 'Pixelate' (Table 11).

4.14 Clean performance

Since FOCUS is primarily an image denoising approach, we wanted to evaluate performance on clean images (Table 12). For the model adaptation methods, we append an additional category 'Clean' (comprising the original test images) to the existing 15 corruptions for ADE20k-C and NYU2k-C. For semantic segmentation, FOCUS demonstrated stronger performance compared to 'TENT', 'CoTTA' and 'DUSA' while showing similar clean performance with 'DDA'. For depth estimation, FOCUS demonstrated competitive performance with CoTTA. The results suggest that FOCUS preserves the much of the original semantic information for both semantic segmentation and depth estimation with minimal degradation, compared to model adaptation methods, which do not effectively adapt to the clean images and restore original performance.

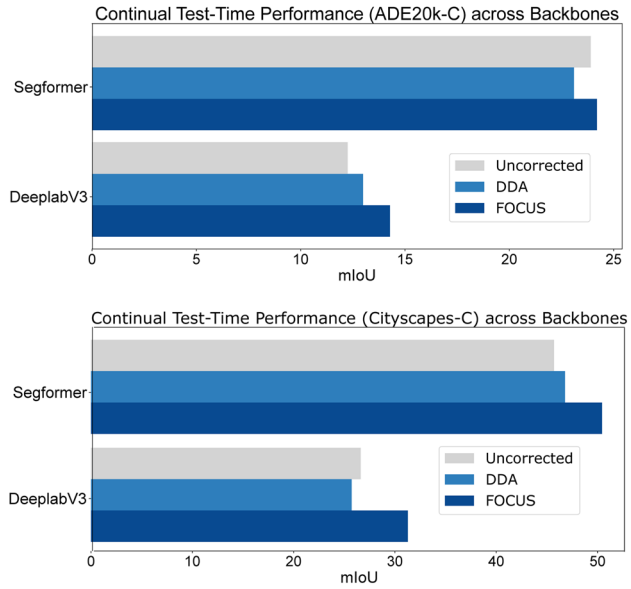

Fig. 10 Performance across transformer-based (Segformer) and convolutional-based (DeeplabV3-R101) architectures for ADE20k-C and Cityscapes-C

4.15 Performance on convolutional and transformer based architectures

FOCUS demonstrates consistent improvements over DDA and baseline uncorrected performance for both transformer and CNN-based architectures (Fig. 10), suggesting that our learned frequency priors are generalizable.

4.16 Limitations

While the Y-FPN layer itself is lightweight (with fast inference) and requires modest computational resources during training, FOCUS has longer inference times due to diffusion-driven denoising, which may limit its applicability for

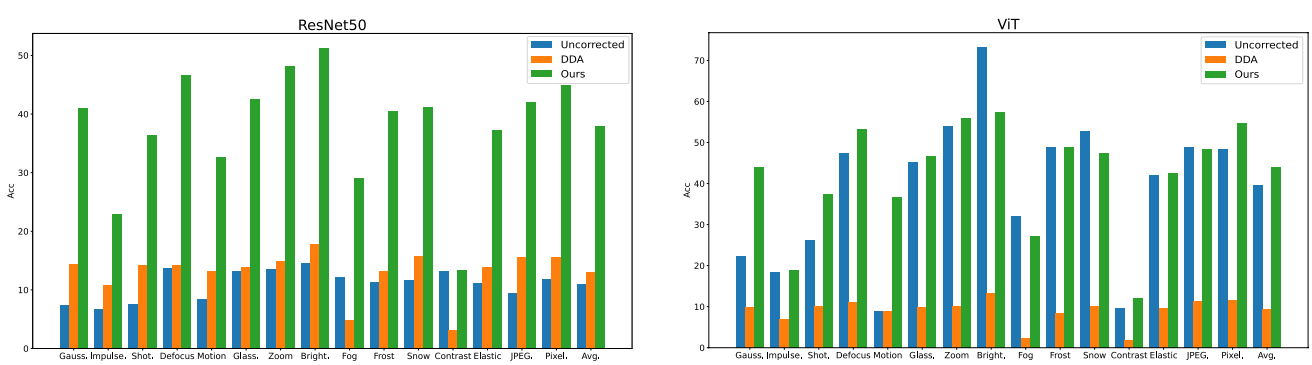

Fig. 11 Image Classification (CIFAR-100c) for ResNet50 and ViT. For our approach, we condition the denoising process with the learned low-frequency filters to determine the effectiveness of spatially adaptive filters on image classification

real-time test-time adaptation. To address this, we explored sparse, intermittent supervision using pseudo labels derived from FOCUS-denoised images (Sect. 4.7).

Our experiments demonstrate that intermittent regularization with denoised images improves adaptation performance, FOCUS synergizes effectively with model-adaptation methods. This suggests that combining both approaches maintains model plasticity by adapting to test data while preserving critical task-relevant information through fine-tuning with denoised inputs.

5 Conclusion

In this work, we introduce Frequency Optimized Conditioning of DiffUSion (FOCUS), a frequency-aware, diffusion-driven input adaptation framework for continual test-time adaptation in dense prediction. FOCUS conditions the denoising process on frequency priors extracted by a lightweight Y-shaped Frequency Prediction Network (Y-FPN) that is trained via a novel FrequencyMix augmentation strategy. Beyond strong standalone gains across diverse datasets and tasks, FOCUS enhances existing model adaptation methods by generating reliable pseudo-labels from the denoised inputs, which also mitigates catastrophic forgetting. We believe this work opens new directions for frequency-aware conditioning of pretrained generative models to improve robustness and adaptability in dense prediction under real-world distribution shifts, particularly for complex corruptions with multiple causes [66].

Appendix

Image classification

We also evaluate performance for FOCUS on CIFAR-100c [67] (Fig. 11). FOCUS outperforms DDA across all corruptions for both ResNet50 and ViT, highlighting the effectiveness of a learned conditioning approach during diffusion-driven denoising for image classification tasks.

Quantitative results for semantic segmentation and depth estimation

We include the full quantitative results for test-time adaptation performance for semantic segmentation (ADE20k-C and Cityscapes-C) and depth estimation (NYU2k-C) in Tables 13, 14, 15 and 16.

Table 13 Test time adaptation performance on Cityscapes-C (semantic segmentation)

Time	*t*															
Method	Noise			Blur				Weather				Digital				Avg. mIoU (↑)
	Gauss	Shot	Impul	Defoc	Glass	Motion	Zoom	Snow	Frost	Fog	Brit	Contr	Elastic	Pixel	JPEG	
Segformer	15.1	17.0	16.7	*56.7*	44.3	56.0	24.6	25.7	28.5	**72.7**	*78.6*	**66.9**	*77.8*	68.0	37.1	45.7 ± 23.7
Tent (ICLR'21)	17.5	19.7	17.5	**56.8**	45.5	56.0	**25.0**	27.0	**30.5**	70.2	77.3	64.8	76.6	65.4	30.7	45.3 ± 22.4
CoTTA (CVPR'22)	19.2	23.7	21.4	56.5	44.9	*56.5*	24.6	27.1	28.2	*72.2*	**79.0**	*65.9*	**78.2**	67.3	36.2	46.7 ± 22.3
DDA	*34.4*	*42.5*	*33.1*	54.2	**52.2**	53.2	20.0	*32.8*	16.1	60.3	76.1	41.1	74.4	*72.4*	39.0	46.8 ± 18.7
FOCUS (Ours)	**42.0**	**51.8**	**42.8**	55.1	*50.2*	55.2	21.5	**33.5**	17.7	67.1	75.1	53.2	76.4	**74.1**	*40.9*	**50.4** ± 18.1
FOCUS (Ours) + TENT (*k*=1)	19.3	25.4	25.6	56.2	49.7	**56.8**	*24.7*	32.6	*29.5*	71.0	78.4	57.8	77.7	69.9	**42.7**	*47.8* ± 20.8

Bold and *Italics* denote best and runner-up results. We compute standard deviation across corruptions and evaluate performance for model adaptation with additional intermittent supervision (every *k* iterations) with pseudo labels from the FOCUS denoised images

Table 14 Test time adaptation performance on ADE20k-C (semantic segmentation)

Time	*t*															
Method	Noise			Blur				Weather				Digital				Avg. mIoU (↑)
	Gauss	Shot	Impul	Defoc	Glass	Motion	Zoom	Snow	Frost	Fog	Brit	Contr	Elastic	Pixel	JPEG	
Segformer	14.2	15.8	15.6	**23.1**	16.8	**22.5**	**10.3**	**22.3**	**21.5**	**38.6**	**42.0**	23.1	**24.5**	33.1	35.3	*23.9* ± 9.1
Tent (ICLR'21)	11.2	13.6	13.6	16.8	14.0	17.0	6.8	17.5	17.0	27.5	30.5	15.7	18.2	22.4	22.9	17.6 ± 5.9
CoTTA (CVPR'22)	14.6	15.7	14.7	21.9	15.2	20.6	8.5	18.7	17.3	34.8	38.9	15.0	20.5	27.8	28.4	20.8 ± 8.0
DDA (CVPR'23)	16.5	16.8	16.3	*22.7*	*18.4*	20.5	9.1	19.6	18.6	32.7	39.4	19.8	*24.4*	**36.1**	*35.7*	23.1 ± 8.5
DUSA (NeurIPS'24)	**21.2**	**24.0**	**22.7**	21.7	**24.0**	20.2	7.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.4 ± 10.7
FOCUS (Ours)	17.1	18.0	17.7	22.5	17.3	*21.9*	*10.1*	*20.1*	*20.3*	*38.3*	41.6	24.6	24.3	*35.0*	*35.7*	24.3 ± 8.8
FOCUS (Ours) + DUSA (*k*=1)	18.5	*20.7*	*19.4*	22.2	*18.4*	*21.9*	*10.1*	19.4	19.3	37.5	*41.8*	**25.9**	24.1	**36.1**	**36.4**	**24.8** ± 8.7
FOCUS (Ours) + DUSA (*k*=5)	*19.2*	19.9	18.2	21.0	17.4	21.1	9.6	19.7	18.7	37.0	41.0	*25.6*	22.3	34.6	35.3	24.0 ± 8.5

Bold and *Italics* denote best and runner-up results. We compute standard deviation across corruptions and evaluate performance for model adaptation with additional intermittent supervision (every *k* iterations) with pseudo labels from the FOCUS denoised images

Table 15 Test time adaptation performance on NYU2k-C (depth estimation, learning rate 10^{-4})

Time	*t*															
Method	Noise			Blur				Weather				Digital				Avg. Rel. Error ↓
	Gauss	Shot	Impul	Defoc	Glass	Motion	Zoom	Snow	Frost	Fog	Brit	Contr	Elastic	Pixel	JPEG	
DenseDepth	1.27	1.20	1.24	1.04	0.63	*0.65*	*0.56*	0.82	0.91	**0.36**	**0.21**	1.12	**0.28**	**0.15**	0.21	0.71 ± 0.39
Tent (ICLR'21)	0.53	0.47	*0.42*	*0.64*	*0.59*	0.75	0.93	*0.51*	0.61	0.57	0.54	0.93	0.54	0.60	0.75	0.63 ± 0.15
DDA (CVPR'23)	1.30	1.21	1.29	1.08	0.78	0.81	0.88	0.82	0.78	0.77	*0.23*	1.40	0.27	0.15	*0.19*	0.80 ± 0.41
FOCUS (Ours)	**0.22**	**0.24**	**0.19**	1.07	0.70	0.71	0.70	0.71	**0.38**	0.55	*0.23*	*0.88*	0.29	*0.16*	**0.17**	*0.48* ± 0.29
FOCUS (Ours) + TENT(k=1)	*0.43*	*0.38*	0.44	**0.53**	**0.50**	**0.47**	**0.40**	**0.40**	*0.43*	*0.38*	0.31	**0.64**	*0.32*	0.29	0.34	**0.42** ± 0.09

Bold and *Italics* denote best and runner-up results. We compute standard deviation across corruptions and evaluate performance for model adaptation with additional intermittent supervision (every k iterations) with pseudo labels from the FOCUS denoised images. CoTTA diverged to degenerate solutions

Table 16 Test time adaptation performance on NYU2k-C (depth estimation, learning rate 10^{-5}

Time	*t*															
Method	Noise			Blur				Weather				Digital				Avg. Rel. Error ↓
	Gauss	Shot	Impul	Defoc	Glass	Motion	Zoom	Snow	Frost	Fog	Brit	Contr	Elastic	Pixel	JPEG	
DenseDepth	1.27	1.20	1.24	1.04	0.63	0.65	0.56	0.82	0.91	0.36	*0.21*	1.12	0.28	**0.15**	0.21	0.71 ± 0.39
Tent (ICLR'21)	0.50	0.32	0.35	0.50	0.37	0.43	0.40	0.39	0.39	0.33	0.25	0.68	0.25	0.24	0.27	0.38 ± 0.11
CoTTA (CVPR'22)	0.41	0.28	0.36	0.51	*0.29*	*0.33*	*0.30*	*0.37*	0.39	*0.22*	**0.17**	*0.66*	**0.16**	0.18	0.20	*0.32* ± 0.13
DDA (CVPR'23)	1.30	1.21	1.29	1.08	0.78	0.81	0.88	0.82	0.78	0.77	0.23	1.40	0.27	0.15	*0.19*	0.80 ± 0.41
FOCUS (Ours)	**0.22**	*0.24*	**0.19**	1.07	0.70	0.71	0.70	0.71	**0.38**	0.55	0.23	0.88	0.29	*0.16*	**0.17**	0.48 ± 0.29
FOCUS (Ours) + TENT(k=1)	0.41	0.32	0.35	*0.44*	0.38	0.42	0.37	*0.37*	0.39	0.33	0.25	**0.63**	0.25	0.24	0.28	0.36 ± 0.09
FOCUS (Ours) + CoTTA (k=1)	*0.28*	**0.23**	*0.24*	**0.31**	**0.24**	**0.31**	**0.27**	**0.29**	*0.29*	**0.20**	**0.17**	0.71	*0.17*	*0.16*	0.20	**0.27** ± 0.13

Bold and *Italics* denote best and runner-up results. We compute standard deviation across corruptions and evaluate performance for model adaptation with additional intermittent supervision (every k iterations) with pseudo labels from the FOCUS denoised images

Table 17 Study of augmentation depth in FrequencyMix on test-time adaptation performance for Cityscapes-*C* (semantic segmentation)

Time	t															
Depth	Noise			Blur				Weather				Digital				Avg. mIoU (↑)
	Gauss	Shot	Impul	Defoc	Glass	Motion	Zoom	Snow	Frost	Fog	Brit	Contr	Elastic	Pixel	JPEG	
{1,2,3,4}	42.0	51.8	42.8	55.1	50.2	55.2	21.5	33.5	17.7	67.1	75.1	53.2	76.4	74.1	40.9	50.4 ± 18.1
2	35.5	47.7	36.5	53.0	52.5	54.1	19.0	31.4	14.5	62.3	73.0	48.5	74.2	75.8	52.7	48.7 ± 18.8
4	46.3	56.3	46.5	52.9	48.7	54.44	19.8	35.1	18.1	63.6	75.5	35.6	76.0	73.0	43.3	49.7 ± 18.0
8	44.6	54.8	45.5	51.5	49.3	53.1	18.8	34.6	17.7	58.2	74.6	21.3	74.7	74.3	49.8	48.2 ± 18.9

'{1,2,3,4}' refers to the setting where the augmentation depth is randomly drawn at each iteration. We compute standard deviation across corruptions

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Author contributions GT proposed the key idea, designed the workflow, conducted the experiments and drafted the manuscript. ZJ gave valuable feedback on the manuscript and also supervised the research direction of the work. YXL critically reviewed the manuscript and helped to refine the core aspects of the manuscript. XY was involved in the refining the idea and assisted with experimental design. NC helped in revising and writing the manuscript. CXF provided inputs on research and experimentation. IWT provided additional guidance and helped greatly in improving overall manuscript quality through his comments on refining the manuscript's key aspects and contributions. KCK provided regular feedback and gave insights on improving the work. GQ supervised the overall research design and assisted in reviewing the manuscript. All authors participated in the discussions, reviewed the manuscript and approved the final version of the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

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